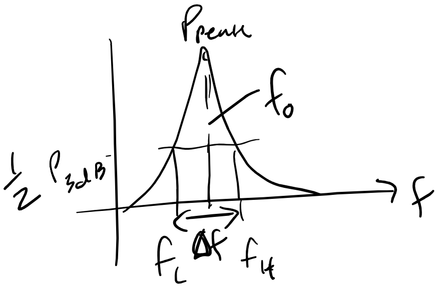


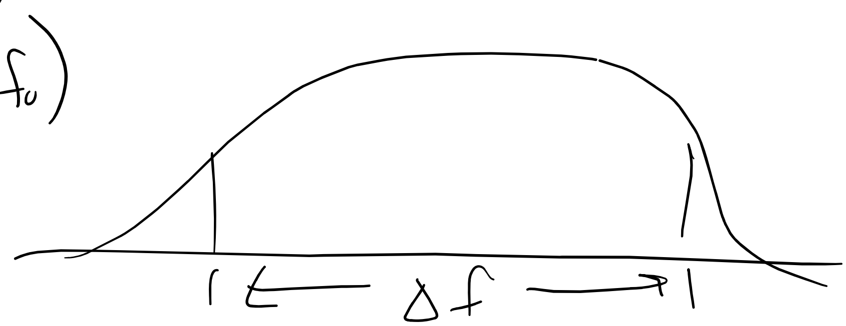
Δf can be used to define the range over which light is coherent

(ℓ) length over which light is coherent

$\ell = c(\Delta t)$

$$\boxed{\ell = \frac{c}{\Delta f}}$$

Sodium Lamp $(568nm \rightarrow f_0)$ → yellow



$\Delta f = 5\times10^{11}$ Hz

$\ell_{coh} = \frac{c}{\Delta f} = \frac{3\times10^{8}}{5\times10^{11}} = 0.6mm$

$\Delta t = \frac{1}{5\times10^{11}} = 2psec$

<u>multimodal Laser (He-Ne)</u> (Gas Laser)

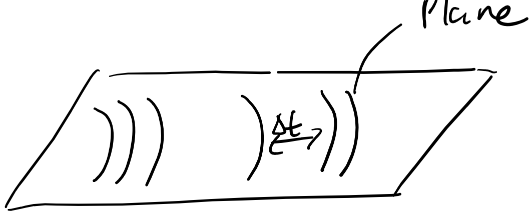
$\Delta f = 1\times10^{9}$ Hz

$\ell_{coh} = 300mm$

<u>Single Mode Laser</u>

$\Delta f \approx 1\times10^{6}$ Hz $\rightarrow \ell_{coh} = 300m$

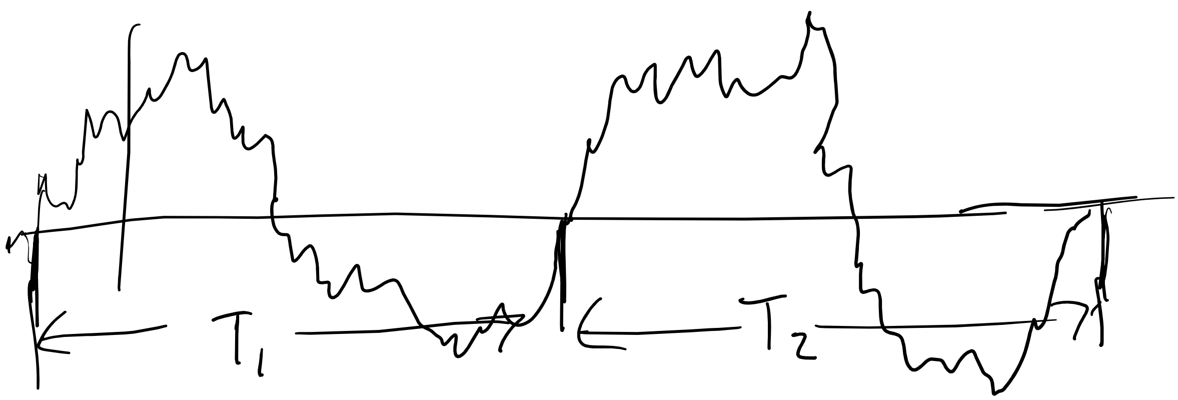
Spatial coherence



- Δt is random
- the shape is well defined

* we could predict wave front shape/magnitudes, could not predict when they happed.

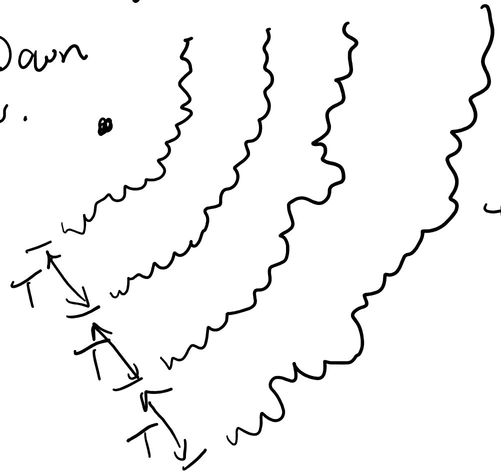
temporal coherence:



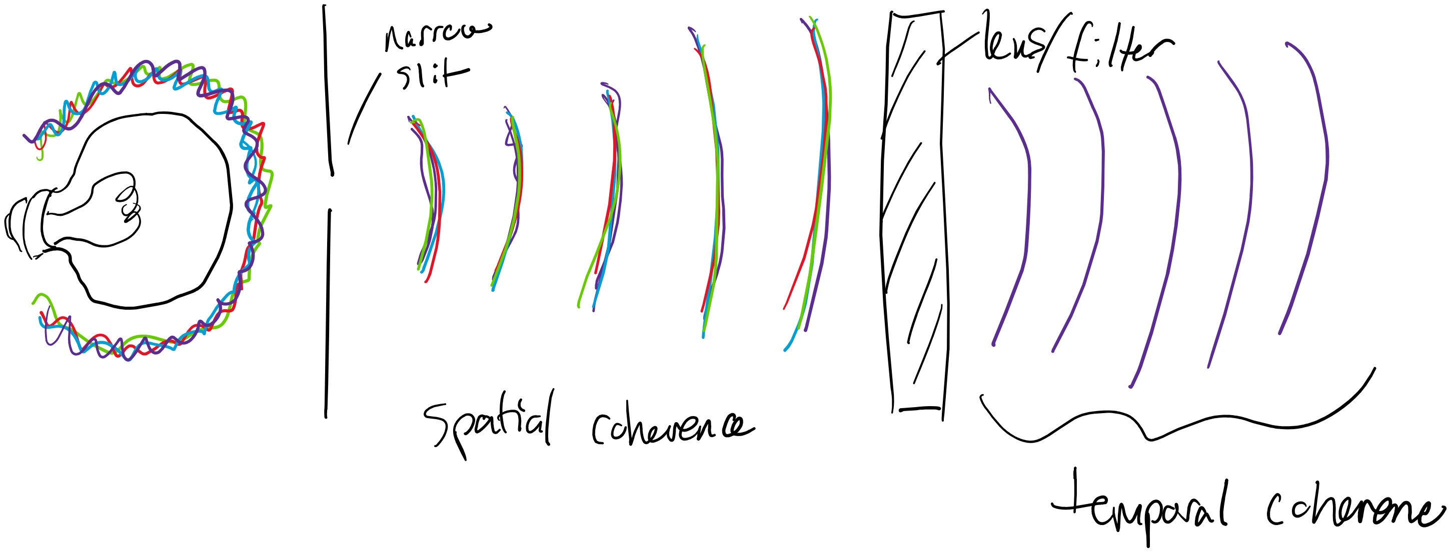
$$T_1 = T_2$$

- can predict the time/phase, cannot predict the shapes or magnitude

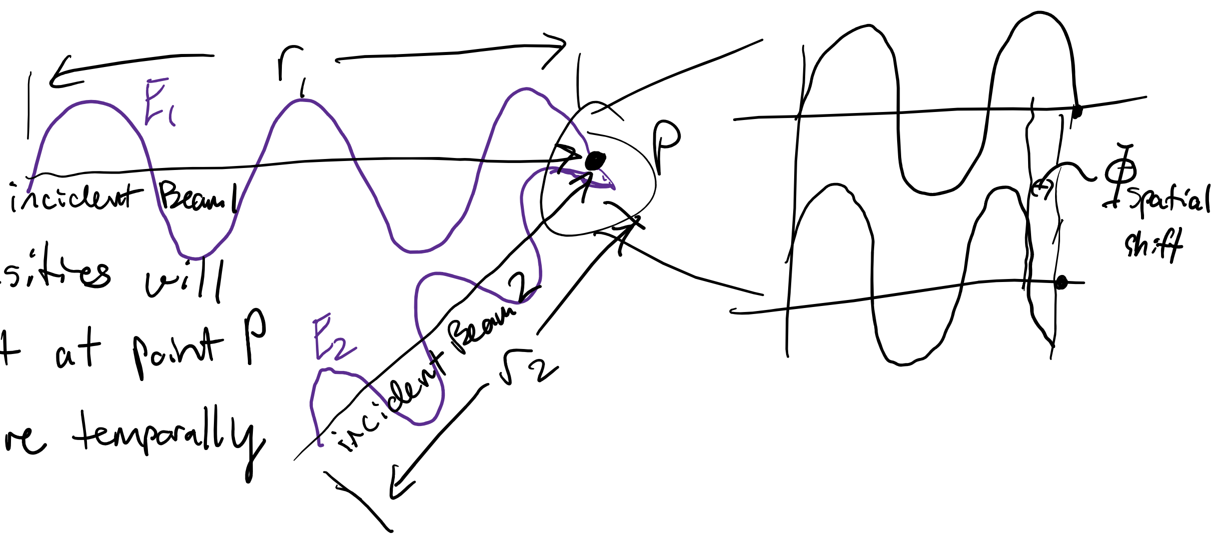
Top Down view.



T is uniform



The importence of coherence is the interference Pattern.



* the intensities will add/subtract at point P b/c they are temporally coherent.

$$\text{Intensity}(I)_{Total} = I_1 + I_2 + 2(I_1 I_2)^{1/2}\cos(\eta)$$

$$\eta = (k(r_2 - r_1)) + (\Phi_2 - \Phi_1)$$

$$\Phi = 0 \rightarrow |E_1 + E_2|$$

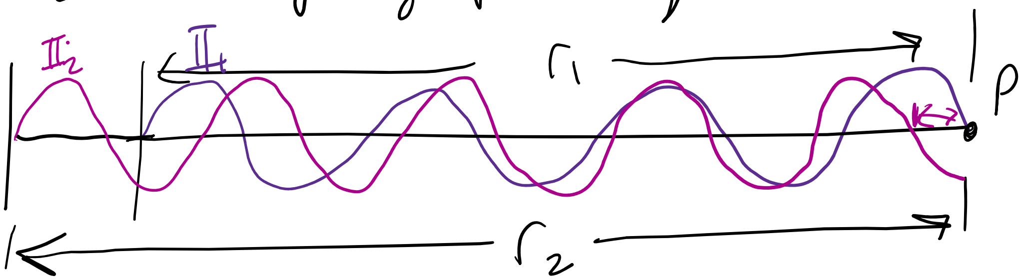
$$\Phi = 180^\circ \rightarrow |E_1 - E_2|$$

$$I_{max} = I_1 + I_2 + 2(I_1 I_2)^{1/2} \cos(0) = 1$$
$$= I_1 + I_2 + 2(I_1 I_2)^{1/2}$$

$$I_{min} = I_1 + I_2 - 2(I_1 I_2)^{1/2} ; \cos(\pi) = -1$$

- when 2 beams travel together (overlapped)

η will vary by "path length"



- changing r_1 and r_2 simply changes the relative wavefront starting position.

<u>constructive interference</u>

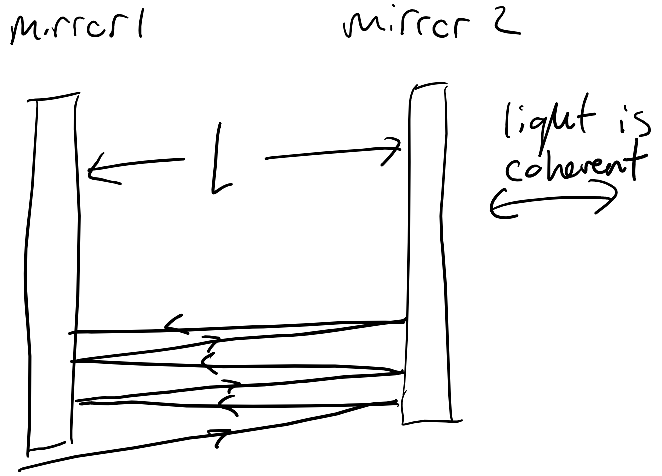
$$k(r_2 - r_1) = 2b\pi ; b \geq 0 \text{ (b is an integer)}$$

even integers

<u>destructive interference</u>

$$k(r_2 - r_1) \approx (2b+1)\pi$$

odd integers



$$m\left(\frac{\lambda}{2}\right) = L$$